

Rose, Blake (GE Research, US)

From: Dai, Jian (GE Research, US)
Sent: Friday, February 17, 2023 12:03 PM
To: Rose, Blake (GE Research, US)
Cc: Szczesny, Paul (GE Research, US)
Subject: Put application in to KR260 QSPI eMMC flash

Hi Blake,

This is how to boot and run your bare metal application.

- Gather all the required files (pmufw.elf, fsbl.elf, < bit file exported by Vivado> and <you application elf file>) in one directory
- Create a bif file (kr260.bif, for ex.) in that directory with the following content:

the_ROM_image:

```
{  
    [fsbl_config] a53_x64  
    [pmufw_image]pmufw.elf  
    [bootloader]fsbl.elf  
    [destination_device=pl]kria_bd_wrapper.bit  
    [destination_cpu=a53-0]a53_perf_test.elf  
}
```

- Run the bootgen command:



```
CA: Command Prompt  
C:\projects\Vitis_Workspace\a53_perf_test\Debug>bootgen -arch zynqmp -image kr260.bif -o boot.bin  
  
***** Xilinx Bootgen v2022.1  
**** Build date : Apr 18 2022-16:02:32  
** Copyright 1986-2022 Xilinx, Inc. All Rights Reserved.  
  
[WARNING]: [fsbl_config] a53_x64 | a53_x32 | r5_single | r5_dual is no more supported. Use 'destination_cpu'  
[INFO] : Bootimage generated successfully  
  
C:\projects\Vitis_Workspace\a53_perf_test\Debug>_
```

- Use the KR260 built-in "Boot Image Recovery Tool" to update the flash
Hold the "FWUEN" button while power on KR260. the serial port will show the IP address of the module. Use web browser to finish the update

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XILINX Boot Image Recovery Tool

System Board Information

Name :	SMK-K26-XCL2G-ED
Revision # :	1
Serial # :	XFL14JKYKHXYX
Part # :	5057-04ED
UUID :	51E33BB5255842A288D7D9945803D7F6

Carrier Card Information

Name :	SCK-KR-G
Revision # :	1
Serial # :	XFL1WMN4PGLE
Part # :	5100-01ED
UUID :	C0ADEEC09F1C46D892906

Boot Image Status

Current Status	Image A : Bootable		
	Image B : Bootable		
	Requested Boot Image : Image B		
	Last Booted Image : Image A		

Image A	Image B	Requested Boot Image
☐ Non Bootable	☐ Non Bootable	☐ Image A
☒ Bootable	☒ Bootable	☒ Image B

[Configure](#)

Recover Image

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Select Image to be recovered

☐ Image A (QSPI)
☒ Image B (QSPI)
☐ Image Wic (eMMC)

[Upload](#)

Upload successful

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- Reboot and your application is going to be running right away.

Jian